

AI-Use Log

Tools and roles. *Claude Code (Anthropic)*: repository work — data acquisition scripts, pipeline code, tests, verification tooling, AI-assisted review of outputs, and this document's prose. *ChatGPT (OpenAI)*: used by the case owner to draft brief interpretation and the structured repair/review instructions given to Claude Code. *Human (case owner)*: accountable for the case, final interpretation and acceptance of every recommendation, and the (still pending) hand labels.

Authority boundaries. AI proposed code, queries, taxonomy logic and opportunity structures. AI could not and did not: create consumer evidence, invent source facts, fill human labels, convert missing evidence into numbers, or grant final acceptance.

Rejected AI suggestion 1 — wrong Amazon category. First attempt streamed the "Appliances" category ("purifiers are appliances"). Returned 32 candidates, dominated by replacement-filter accessories. Rejected after inspection of the actual rows; "Home_and_Kitchen" yielded the real 237 products / 10,547 reviews.

Rejected AI suggestion 2 — first-pass classifier accepted as final. The title+description keyword classifier was proposed as done. Inspection of actual matches found vacuum cleaners and wearable ionizers matched via cross-sell copy in *descriptions*. Rejected; title-only matching plus exclusion regexes produced the frozen 237.

Failed AI check. Q3's classifier first ran without a polarity gate: "Ozone / smell" scored 22% prevalence with a *positive* CSAT impact — "great at eliminating odors" counted as a complaint. Programmatic re-check against raw text exposed it; keywords now only count inside negative-polarity sentences, and the rerun produced the published figures.

Integrity incident (disclosed). An earlier file presented as a hand-labelled sample failed provenance validation — its review texts did not match the real corpus for their review IDs (traced to a synthetic fixture's template pool). It was deleted; regression tests now verify every sample row against the real corpus by id, product and exact text, and Q3 validation remains **blocked pending genuine human labels** — no label was ever substituted.

Verification. AI-derived work was checked by: programmatic checks (59-row evidence table, 301-check verifier, 49-test discovery suite, negative tests that corrupt a trace and expect failure), deterministic re-runs of the offline pipeline, source retrieval against archived pages, and a genuine fresh-clone reproduction (FRESH_CLONE_REPORT.md).